The slope of the normal is the negative of the reciprocal of the tangent which in this case is $\frac{\sin 3t}{\cos 2t}$. Hence, the equation of the normal is

$$y - 3\sin 2t = \frac{\sin 3t}{\cos 2t}(x - 2\cos 3t).$$

That is, $x\sin 3t - y\cos 2t = 2\sin 3t\cos 3t - 3\sin 2t\cos 2t = \sin 6t - \frac{3}{2}\sin 4t$.

## 7.2.5 Angle between two curves

**Definition 7.3**

Angle between two curves, if they intersect, is defined as the acute angle between the tangent lines to those two curves at the point of intersection.

For the given curves, at the point of intersection using the slopes of the tangents, we can measure the acute angle between the two curves. Suppose $y = m_1x + c_1$ and $y = m_2x + c_2$ are two lines, then the acute angle $\theta$ between these lines is given by,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right| \qquad \ldots (3)$$

where $m_1$ and $m_2$ are finite.

**Remark**

(i) If the two curves are parallel at $(x_1, y_1)$, then $m_1 = m_2$.

(ii) If the two curves are perpendicular at $(x_1, y_1)$ and if $m_1$ and $m_2$ exists and finite then $m_1m_2 = -1$.

**Example 7.14**

Find the angle between $y = x^2$ and $y = (x-3)^2$.

**Solution**

Let us now find the point of intersection of the two given curves. Equating $x^2 = (x-3)^2$ we get, $x = \frac{3}{2}$. Therefore, the point of intersection is $\left(\frac{3}{2}, \frac{9}{4}\right)$. Let $\theta$ be the angle between the curves. The slopes of the curves are as follows :

For the curve $y = x^2$,

$$\frac{dy}{dx} = 2x.$$

Let $m_1 = \left.\frac{dy}{dx}\right|_{\text{at } \left(\frac{3}{2}, \frac{9}{4}\right)} = 3.$

For the curve $y = (x-3)^2$,

$$\frac{dy}{dx} = 2(x-3).$$

Let $m_2 = \left.\frac{dy}{dx}\right|_{\text{at } \left(\frac{3}{2}, \frac{9}{4}\right)} = -3.$

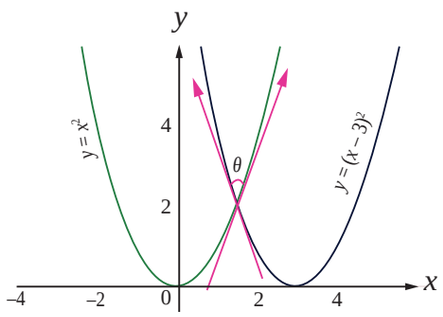



Fig.7.10